

Supplementary Material

S1: Statistical Analysis and Reporting of Experimental Results

Table 1: Experimental performance comparison of different models on various tasks in ADNI, where " _ " denotes the second-best model. Notably, (M) denotes a multimodal method, and # indicates that the corresponding baseline differs significantly from NeurGNN in terms of accuracy according to the exact McNemar test after Holm correction for multiple comparisons.

Tasks	AD vs NC (ADNI)				AD vs MCI (ADNI)				NC vs MCI (ADNI)			
Model	ACC	F1	REC	PRE	ACC	F1	REC	PRE	ACC	F1	REC	PRE
sMRI												
3D ResNet	0.885±0.015 #	0.885±0.014	0.872±0.017	0.897±0.013	0.803±0.021 #	0.803±0.019	0.812±0.023	0.791±0.019	0.681±0.019 #	0.682±0.024	0.690±0.025	0.681±0.022
DA-MIDL	0.862±0.017 #	0.861±0.021	0.852±0.020	0.871±0.017	0.788±0.027 #	0.787±0.025	0.776±0.021	0.797±0.027	0.659±0.027 #	0.662±0.021	0.672±0.029	0.657±0.019
Gao et al.	0.870±0.023 #	0.870±0.017	0.881±0.017	0.863±0.019	0.793±0.021 #	0.793±0.021	0.793±0.023	0.793±0.020	0.669±0.025 #	0.658±0.031	0.648±0.023	0.680±0.034
Segmamba*	0.892±0.021	0.891±0.014	0.878±0.022	<u>0.903</u> ±0.021	0.815±0.023 #	0.815±0.024	0.820±0.027	0.811±0.023	0.706±0.020 #	0.699±0.027	0.682±0.027	0.718±0.025
UNETR*	0.883±0.020 #	0.883±0.019	0.881±0.022	0.885±0.019	0.801±0.025 #	0.801±0.023	0.801±0.023	0.801±0.023	0.702±0.031 #	0.695±0.033	0.678±0.036	0.712±0.031
VT-UNet*	0.888±0.021 #	0.888±0.021	0.892±0.018	0.883±0.023	0.811±0.009 #	0.811±0.016	0.817±0.011	0.805±0.019	0.705±0.023 #	0.703±0.029	0.698±0.027	0.710±0.030
TransBTS*	0.894±0.015	0.894±0.018	0.897±0.017	0.889±0.019	0.819±0.019 #	0.818±0.020	0.815±0.019	0.821±0.022	0.722±0.024	0.714±0.021	0.697±0.021	0.737±0.027
MCADS	0.898±0.023	0.896±0.021	0.900±0.025	0.8929±0.013	0.838±0.024 #	0.835±0.027	<u>0.844</u> ±0.033	0.826±0.027	0.709±0.029 #	0.703±0.027	0.698±0.028	0.708±0.026
GF-Net	0.888±0.019 #	0.886±0.017	0.881±0.021	0.891±0.014	0.811±0.022 #	0.806±0.017	0.810±0.018	0.802±0.017	0.706±0.033 #	0.703±0.031	0.709±0.035	0.697±0.030
MAD-Former	<u>0.900</u> ±0.021	0.897±0.019	0.895±0.018	0.890±0.019	0.840±0.019	0.837±0.020	0.841±0.019	0.832±0.022	0.724±0.027	0.721±0.029	0.725±0.031	0.716±0.030
AAGN	0.882±0.019#	0.881±0.016	0.887±0.017	0.874±0.015	0.819±0.020 #	0.821±0.019	0.825±0.023	0.817±0.017	0.718±0.021 #	0.722±0.020	0.727±0.024	0.717±0.020

dGLCN	0.870±0.011#	0.867±0.012	0.872±0.012	0.861±0.011	0.823±0.017 #	0.823±0.019	0.823±0.022	0.823±0.020	0.712±0.035 #	0.715±0.037	0.720±0.031	0.710±0.035
IMDL(M)	0.887±0.027	0.886±0.022	0.876±0.024	0.896±0.021	<u>0.853</u> ±0.019	<u>0.853</u> ±0.023	0.853 ±0.017	<u>0.853</u> ±0.026	<u>0.755</u> ±0.031	<u>0.751</u> ±0.029	<u>0.739</u> ±0.033	<u>0.766</u> ±0.027
NeurGNN (ours)	0.912 ±0.002	0.912 ±0.002	0.912 ±0.024	0.912 ±0.024	0.869 ±0.009	0.865 ±0.008	0.841±0.020	0.891 ±0.029	0.756 ±0.024	0.753 ±0.014	0.740 ±0.017	0.767 ±0.049
NeurGNN (CI)	[0.9086, 0.9154]	[0.9094, 0.9146]	[0.8819, 0.9421]	[0.8824, 0.9424]	[0.8573, 0.8807]	[0.8559, 0.8757]	[0.8167, 0.8665]	[0.8551, 0.9277]	[0.7256, 0.7868]	[0.7347, 0.7713]	[0.7195, 0.7617]	[0.7060, 0.8296]
PET												
3D ResNet	0.890±0.037 #	0.888±0.031	0.875±0.033	0.903±0.029	0.841±0.032 #	0.840±0.033	0.826±0.031	<u>0.859</u> ±0.033	0.666±0.041 #	0.699±0.040	<u>0.776</u> ±0.040	0.650±0.040
PCC-GAN	0.911±0.042	0.908±0.032	0.880±0.037	0.940 ±0.029	0.840±0.027 #	0.838±0.028	0.836±0.026	0.844±0.029	0.684±0.033 #	0.678±0.037	0.663±0.039	0.708±0.035
M3T	0.888±0.050	0.893±0.041	<u>0.932</u> ±0.036	0.860±0.047	0.838±0.036 #	0.845±0.025	0.880 ±0.027	0.813±0.023	0.728±0.037	0.723±0.034	0.725±0.037	0.740±0.033
AmyloidPETNet	0.859±0.041 #	0.859±0.039	0.859±0.043	0.862±0.038	0.816±0.039 #	0.813±0.037	0.802±0.040	0.831±0.037	0.611±0.041 #	0.643±0.040	0.765±0.039	0.600±0.040
SegMamba*	0.903±0.048	0.901±0.040	0.890±0.040	0.914±0.039	0.797±0.048 #	0.787±0.041	0.755±0.040	0.829±0.042	0.670±0.042 #	0.706±0.035	0.797 ±0.033	0.643±0.037
VT-UNet*	<u>0.919</u> ±0.027	0.919 ±0.031	0.922±0.029	<u>0.921</u> ±0.033	0.852±0.017	<u>0.846</u> ±0.015	0.819±0.020	0.878 ±0.015	<u>0.751</u> ±0.038	<u>0.743</u> ±0.037	0.761±0.026	<u>0.765</u> ±0.035
TransBTS*	0.899±0.024 #	0.898±0.027	0.893±0.027	0.907±0.026	0.836±0.018 #	0.837±0.019	0.845±0.019	0.834±0.019	0.699±0.044 #	0.706±0.044	0.733±0.049	0.709±0.039
MCADS	0.916±0.035	0.909±0.037	0.902±0.037	0.895±0.036	0.842±0.039 #	0.839±0.037	0.834±0.037	0.844±0.037	0.731±0.042	0.688±0.040	0.702±0.041	0.674±0.040
GF-Net	0.894±0.025 #	0.891±0.029	0.887±0.021	0.894±0.031	0.842±0.041 #	0.841±0.039	0.843±0.037	0.839±0.040	0.718±0.033 #	0.715±0.034	0.720±0.027	0.709±0.037
MAD-Former	0.916±0.027	<u>0.917</u> ±0.025	0.921±0.024	0.913±0.025	0.848±0.037 #	<u>0.846</u> ±0.035	0.846±0.038	0.846±0.032	0.731±0.031	0.733±0.031	0.737±0.028	0.729±0.029
AAGN	0.888±0.031 #	0.887±0.027	0.885±0.028	0.889±0.027	0.838±0.031 #	0.839±0.033	0.841±0.027	0.837±0.035	0.726±0.039	0.727±0.036	0.723±0.035	0.717±0.036
dGLCN	0.906±0.033	0.906±0.031	0.906±0.032	0.906±0.028	0.841±0.034 #	0.840±0.034	0.842±0.031	0.837±0.034	0.718±0.051	0.717±0.049	0.715±0.047	0.719±0.044
IMDL(M)	0.887±0.037 #	0.886±0.035	0.876±0.033	0.896±0.036	0.853±0.028	0.853 ±0.031	<u>0.853</u> ±0.027	0.853±0.030	0.755 ±0.037	0.751 ±0.036	0.739±0.035	0.766 ±0.036
NeurGNN (ours)	0.936 ±0.008	0.916±0.017	0.935 ±0.031	0.899±0.018	0.866 ±0.026	0.839±0.033	0.822±0.038	0.858±0.048	0.738±0.017	0.664±0.014	0.644±0.014	0.690±0.017

NeurGNN (CI)	[0.9249, 0.9471]	[0.8957, 0.9379]	[0.8971, 0.9745]	[0.8763, 0.9221]	[0.8341, 0.8995]	[0.7978, 0.8802]	[0.7741, 0.8703]	[0.7979, 0.9181]	[0.7160, 0.7600]	[0.6460, 0.6820]	[0.6263, 0.6617]	[0.6687, 0.7113]
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Table 2: Experimental performance comparison of different models on various tasks in OASIS, where " _ " denotes the second-best model.

Tasks	AD vs NC (OASIS)			
Model	ACC	F1	REC	PRE
SMRI				
3D ResNet	0.794±0.035 #	0.809±0.051	<u>0.873</u> ±0.049	0.823±0.053
DA-MIDL	0.750±0.057 #	0.739±0.053	0.708±0.056	0.772±0.051
Gao et al.	0.760±0.047 #	0.752±0.044	0.729±0.041	0.777±0.045
Segmamba*	0.800±0.061	0.804±0.063	0.830±0.060	<u>0.850</u> ±0.065
UNETR*	0.791±0.057	0.795±0.051	0.812±0.056	0.780±0.050
VT-UNet*	0.798±0.043	0.798±0.041	0.808±0.041	0.810±0.042
TransBTS*	<u>0.803</u> ±0.041	<u>0.813</u> ±0.039	0.858±0.046	0.774±0.035
MCADS	0.798±0.051	0.796±0.051	0.802±0.050	0.787±0.050
GF-Net	0.785±0.034	0.787±0.031	0.800±0.029	0.772±0.030
MAD-Former	0.800±0.037 #	0.805±0.034	0.825±0.033	0.782±0.038
AAGN	0.784±0.045 #	0.787±0.047	0.812±0.042	0.762±0.044
dGLCN	0.798±0.048	0.791±0.049	0.808±0.047	0.773±0.051
IMDL(M)	0.802±0.053	0.781±0.051	0.708±0.049	0.871 ±0.049

NeurGNN (ours)	0.809 ±0.031	0.838 ±0.033	0.889 ±0.037	0.814±0.039
NeurGNN (CI)	[0.7705, 0.8475]	[0.7970, 0.8790]	[0.8431, 0.9349]	[0.7656, 0.8624]
PET				
3D ResNet	0.828±0.041 #	0.825±0.047	0.809±0.045	0.847±0.047
PCC-GAN	0.801±0.034 #	0.792±0.040	0.777±0.035	0.821±0.042
M3T	0.845±0.054	0.851±0.047	0.877±0.045	0.833±0.045
AmyloidPETNet	0.716±0.088 #	0.757±0.092	<u>0.878</u> ±0.091	0.669±0.089
SegMamba*	0.784±0.079 #	0.759±0.069	0.704±0.075	<u>0.859</u> ±0.067
VT-UNet*	0.785±0.094	0.759±0.081	0.709±0.087	0.845±0.079
TransBTS*	0.853 ±0.054	0.863 ±0.049	0.928 ±0.047	0.821±0.050
MCADS	0.821±0.057 #	0.819±0.051	0.843±0.051	0.798±0.050
GF-Net	0.793±0.041 #	0.798±0.045	0.781±0.044	0.814±0.039
MAD-Former	0.835±0.043	0.827±0.038	0.840±0.042	0.813±0.037
AAGN	0.801±0.039 #	0.795±0.041	0.771±0.033	0.820±0.043
dGLCN	0.812±0.051 #	0.807±0.049	0.819±0.044	0.792±0.046
IMDL(M)	0.802±0.047 #	0.781±0.045	0.708±0.038	0.871 ±0.047
NeurGNN (ours)	<u>0.849</u> ±0.041	<u>0.853</u> ±0.039	<u>0.878</u> ±0.043	0.837±0.044
NeurGNN (CI)	[0.7982, 0.9010]	[0.8047, 0.9029]	[0.825, 0.931]	[0.782, 0.892]

S2: Theoretical and Quantitative Evaluation of BRGC

We added quantitative analyses of intra class dispersion S_W , inter class center distance S_B , and their ratio R_{geo} .

$$S_W = \frac{1}{B} \sum_i \|z_i - c_{y_i}\|_2^2 \quad (1)$$

$$S_B = \frac{2}{K(K-1)} \sum_{k < l} \|c_k - c_l\|_2^2 \quad (2)$$

$$R_{\text{geo}} = \frac{S_B}{S_W + \epsilon} \quad (3)$$

As shown in **Table 3**, incorporating BRGC reduced S_W from 0.427 to 0.394, indicating more compact intra class distributions. S_B increased from 0.819 to 0.837, indicating greater separation between class centers. The increase in R_{geo} further demonstrates improved overall geometric discriminability.

we additionally conducted comparative experiments with different geometric regularization objectives. Specifically, while keeping the NeurGNN architecture, data partitioning, optimization strategy, and cross-entropy supervision unchanged, we separately introduced a ratio-based objective, a margin-based objective, a supervised contrastive objective, and BRGC as auxiliary constraints. We then compared their effects on the intra-class dispersion S_W , inter-class center distance S_B , and their ratio R_{geo} .

Specifically, the ratio-based objective uses the ratio between intra-class dispersion and inter-class center distance as an auxiliary constraint, defined as $L_{\text{ratio}} = \frac{L_{\text{intra}}}{L_{\text{inter}} + \epsilon}$, where minimizing this ratio encourages both reduced intra-class dispersion and increased inter-class separation. The margin-based objective introduces a fixed margin m between intra-class dispersion and inter-class distance and is defined as $L_{\text{margin}} = \max(0, L_{\text{intra}} - L_{\text{inter}} + m)$. An additional penalty is imposed only when the difference between the two terms does not satisfy the predefined margin. The supervised contrastive objective constructs positive and negative sample

pairs according to class labels, encouraging representations of samples from the same class to become closer while separating those from different classes in the normalized feature space, thereby enhancing intra-class consistency and inter-class separability.

Compared with the setting without geometric regularization, all auxiliary objectives improved the geometry of the feature space to varying degrees. Among them, the contrastive objective achieved the lowest S_W , indicating a certain advantage in promoting intra-class compactness. In contrast, BRGC achieved the largest S_B and the highest R_{geo} , suggesting a more balanced effect on preserving intra-class compactness while enhancing inter-class center separation. These results further support that BRGC can, to some extent, regulate the relative geometric relationship between intra-class dispersion and inter-class separation.

Table 3: Quantitative distance analysis results.

Configuration	S_W	S_B	R_{geo}
None	0.427	0.819	1.918
Ratio-based	0.409	0.835	2.057
Margin-based	0.399	0.825	2.070
Contrastive	0.387	0.829	2.118
BRGC	0.394	0.837	2.124

S3: Model efficiency analysis

We added a model efficiency analysis that systematically reports computational complexity, FLOPs, total parameter count, overall training time, average inference time per sample, and GPU memory consumption for different methods. As shown in Figure 4, all experiments were conducted under an identical hardware configuration using a single NVIDIA RTX 3090 GPU with 24 GB of memory. The batch size was fixed at 8 for all models to ensure a fair comparison, and both inference time and GPU memory consumption were measured under the same testing conditions. We selected several representative deep learning models for comparison, including CNN-based, Transformer-based, Mamba-based, and graph neural network-based approaches. SegMamba exhibited relatively high parameter counts and GPU memory consumption, mainly because it employs a complete 3D encoder comprising multistage high-dimensional feature extraction modules and Mamba sequence modeling units.

For NeurGNN, although the graph construction process involves KNN-based pairwise node relationship computation with a theoretical complexity of $O(N^2)$, where N denotes the number of valid patch nodes retained after BSA, BSA effectively removes a substantial number of background-dominated patches before graph construction, thereby markedly reducing the number of nodes involved in relational computation. Consequently, the theoretical quadratic complexity does not directly translate into a high practical computational burden. Experimental results show that NeurGNN requires only 1517.36 s for training, the lowest among the compared methods, while its GPU memory consumption is only 2530 MB, also demonstrating a clear advantage. Its computational cost is 44.098 GFLOPs, remaining at a moderate level overall. These results indicate that, while incorporating brain-region-aware graph construction and relational reasoning, NeurGNN maintains favorable practical computational efficiency and relatively low hardware resource requirements.

Table 4: Model efficiency analysis.

Model	Total Training Time (s)	Average Inference Time per Sample (ms)	GPU Memory (MB)	Total Parameters (M)	FLOPs (G)	Computational complexity
3D_ResNet	5332.05	10.19	6730	63.511	83.541	$O(N)$
PCC_GAN	3656.95	3.86	5252	11.732	5.894	$O(N)$
SegMamba	22935.17	55.78	13622	80.184	87.146	$O(N)$
TransBTS	2939.47	3.21	7416	61.261	18.553	$O(N^2)$
VT_UNet	13294.40	59.98	22870	21.186	40.553	$O(N^2)$
dGLCN	10500.28	22.54	3571	10.241	46.47	$O(N)$
NeurGNN	1517.36	40.17	2530	80.696	44.098	$O(N^2)$

S4: Interpretability validation

To further strengthen the interpretability validation, we supplemented three additional experiments:

1) a random-edge control experiment; 2) a quantitative stage-wise brain-region preservation analysis; and 3) a cross-fold stability analysis.

1) Random-edge control experiment. To further investigate whether the predictive

contribution of the L.mAmyg-associated edges is specific rather than merely a consequence of perturbing arbitrary graph connections, we extended the edge-masking analysis in Table V by incorporating a random-edge control experiment. Specifically, while keeping the model parameters fixed and without retraining, we used the ADNI sMRI AD vs NC classification task and the Stage 2 graph structure. We first excluded all edges directly connected to L.mAmyg and then randomly selected 100 edges from the remaining graph connections as the random-edge controls.

For each random edge e_r , we applied exactly the same procedure as in the original L.mAmyg-associated edge-masking experiment: only the selected edge was removed, while all other node features and graph connections remained unchanged, and the classification accuracy was recalculated. The contribution of each random edge was defined as:

$$\Delta Acc_r = Acc_{\text{full}} - Acc_r^-. \quad (4)$$

Subsequently, we calculated the mean accuracy drop for the L.mAmyg-associated edges and the random edges, respectively:

$$\overline{\Delta Acc} = \frac{1}{M} \sum_{i=1}^M \Delta Acc_i, \quad (5)$$

The experimental results are presented in **Table 5**. The L.mAmyg-associated edges yielded a larger $\overline{\Delta Acc}$ than the randomly selected graph connections, indicating that masking the L.mAmyg-related connections identified by the model resulted in a more pronounced performance degradation. This finding further suggests that the key brain-region associations identified by NeurGNN are not merely attributable to arbitrary graph-edge perturbations, but instead make a more substantial contribution to the model’s predictive performance.

Table 5. Performance analysis of the random-edge control experiment.

Edge type	Number of edges	$\overline{\Delta Acc}$
L.mAmyg-associated edges	8	0.00625
Random edges	100	0.00245

2) Quantitative stage-wise brain-region preservation analysis. We further added a quantitative analysis based on all brain-region labels in the supplementary material. Specifically, we identified

all brain-region labels retained at Stages 1–4 and calculated the number of regions shared between consecutive stages, together with their retention rates relative to the preceding stage. As shown in **Table 6**, 246 brain regions were retained at Stage 1 and 241 at Stage 2, with 241 regions shared between the two stages, corresponding to a retention rate of 97.97%. From Stage 2 to Stage 3, 160 of the 241 regions remained represented in the subsequent stage, yielding a retention rate of 66.39%. From Stage 3 to Stage 4, 34 of the 161 regions remained represented, corresponding to a retention rate of 21.12%. The brain-region retention rate was relatively high from Stage 1 to Stage 2. As the number of downsampling stages increased, the number of retained brain regions gradually decreased, consistent with the progressive reduction in spatial resolution and hierarchical feature integration within the network. The R.cHipp example in the main manuscript is primarily used to illustrate changes in brain-region associations across different stages, whereas the additional whole-brain quantitative analysis provides an overall assessment of the actual brain-region retention at each stage, thereby avoiding reliance on a single brain region for the stage-wise analysis.

Table 6. Quantitative analysis of stage-wise brain-region retention.

Stage comparison	Regions in previous stage	Regions in later stage	Common regions	Retention rate
S1 → S2	246	241	241	97.97%
S2 → S3	241	161	160	66.39%
S3 → S4	161	37	34	21.12%

3) Cross-fold stability analysis. Consistent with the interpretability analysis presented in the main manuscript, we continued to use L.mAmyg as the central node and evaluated the recurrence of its eight associated brain regions in the AD and NC groups across the five cross-validation folds. For each associated brain region, an occurrence was recorded if it was again identified as being associated with L.mAmyg in a given fold, and its frequency of occurrence across the five folds was subsequently calculated. The experimental results are presented in **Tables 7 and 8**. In the AD group, the eight L.mAmyg-associated brain regions showed an average occurrence frequency of 4.13/5 (82.5%) across the five folds. Among them, L.vIa and R.TI were consistently identified in all five folds, while all other regions were identified in at least three folds. In the NC group, the eight associated brain regions showed an average occurrence frequency of 3.75/5 (75.0%), with all

regions likewise recurring in at least three folds. Across the AD and NC groups, the overall recurrence rate of the associated brain regions was 78.75%. These results indicate that, despite variations in the training samples across different cross-validation folds, the major brain-region associations involving L.mAmyg were repeatedly identified across different data partitions. This finding suggests that the principal interpretability results of NeurGNN are not dependent on a specific fold and exhibit good cross-fold consistency.

Table 7: Cross-fold occurrence frequency of L.mAmyg-associated brain regions in AD.

L.mAmyg	L.vla	L.rTtha	L.rHipp	L.cLinG	R.mAmyg	R.A37mv	R.TI	R.cLinG
Fold 1	√	√	√	√	√	√	√	√
Fold 2	√	√	√	√	√	√	√	√
Fold 3	√	√	√	√	√	√	√	√
Fold 4	√		√			√	√	√
Fold 5	√	√			√		√	
Frequency	5	4	4	3	4	4	5	4

Table 8: Cross-fold occurrence frequency of L.mAmyg-associated brain regions in NC.

L.mAmyg	L.vla	L.rTtha	R.cHipp	L.cLinG	L.A20r	L.lAmyg	R.dCa	L.rLinG
Fold 1	√	√	√	√	√	√	√	√
Fold 2	√		√	√	√		√	√
Fold 3	√	√	√	√		√		√
Fold 4	√	√	√		√	√	√	√
Fold 5		√				√	√	
Frequency	4	4	4	3	3	4	4	4